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Department of Mathematics, Bioinformatics and Computer Application

HOSPITAL INFORMATION SYSTEM

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INTRODUCTION

- **Integrated Digital Platform:** A comprehensive **Hospital Information System (HIS)** designed to digitalize clinical, administrative, and financial operations.
- **Hierarchical Staff Management:** Utilizes an **Employee Superclass** model to efficiently organize records for Admins, Doctors, and Medical Staff.
- **End-to-End Patient Workflow:** Tracks the complete patient journey, including registration, appointments, diagnosis, and treatment.
- **Automated Clinical Operations:** Digitally links doctor prescriptions directly to **Pharmacy Inventory** and **Lab Reports** for real-time accuracy.
- **Smart Resource Allocation:** Provides real-time tracking of room availability and nursing staff assignments with automated workload balancing.
- **Consolidated Billing Module:** A transparent financial system that auto-calculates charges for consultations, laboratory tests, medications, and room stays.

SCOPE

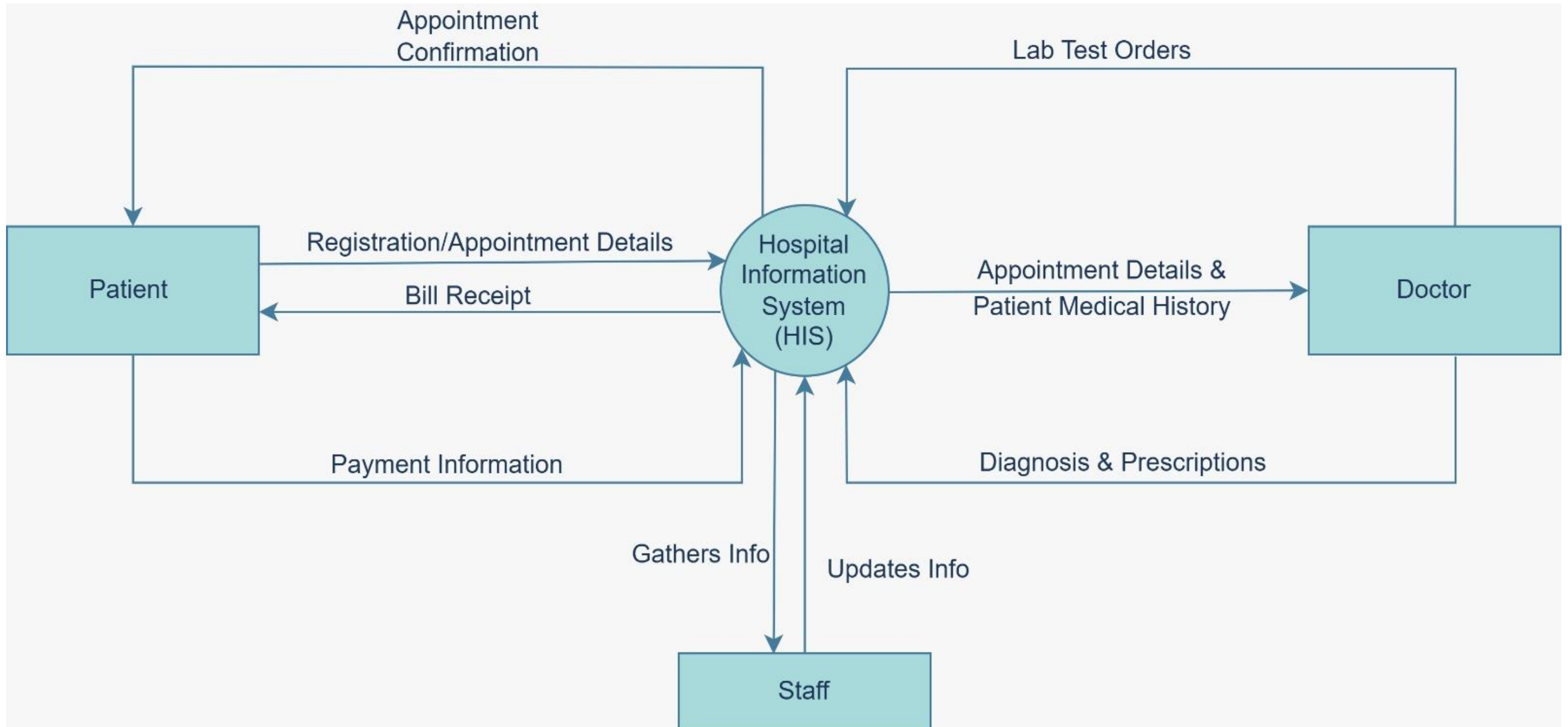


- **Patient Administration:** Patient registration, Unique ID, digital records.
- **Electronic Medical Records (EMR):** Store history, reports, prescriptions with secure access.
- **Doctor Management:** Manage doctor details, schedule, and role-based access.
- **Appointment Scheduling:** Online/offline booking with reminder system.
- **Laboratory Management:** Record tests and link lab reports to patient records.
- **Pharmacy Management:** Medicine inventory, digital prescriptions, stock alerts.
- **Billing Management:** Automatic invoice generation and payment tracking.
- **Bed & Ward Management:** Track room availability and patient admission.
- **Data Security & Backup:** Secure database, encryption, automated backup.
- **Scalability & Integration:** Multi-branch support and future-ready cloud system.

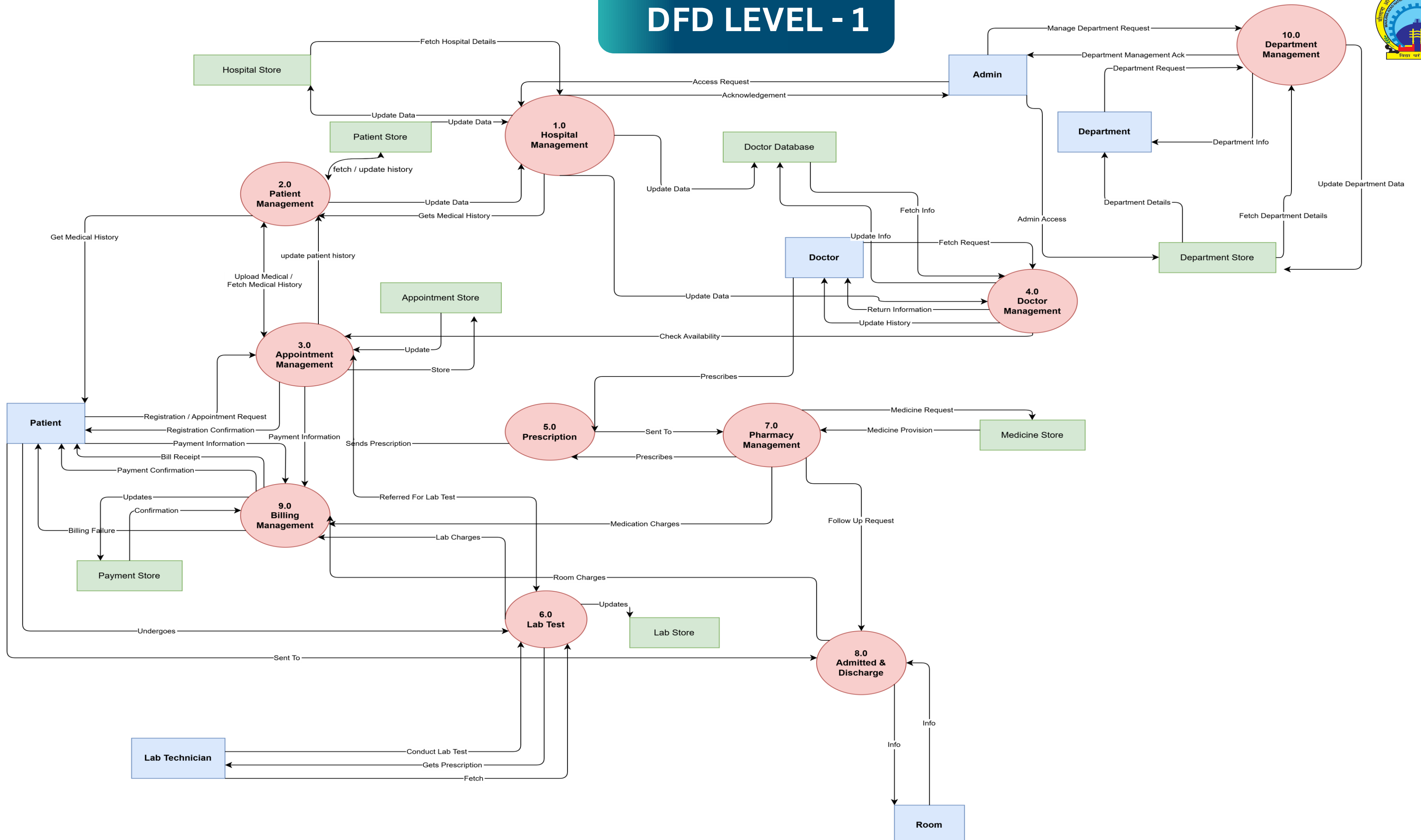
OBJECTIVES

1. Digital Patient Records.
2. Emergency Response Optimization.
3. Inpatient Care Management
4. Data Security & Backup.
5. Operational Automation.
6. Centralized Doctor Management.
7. Modular Billing System.
8. Laboratory & Pharmacy Integration.
9. Staff Workload Balancing.

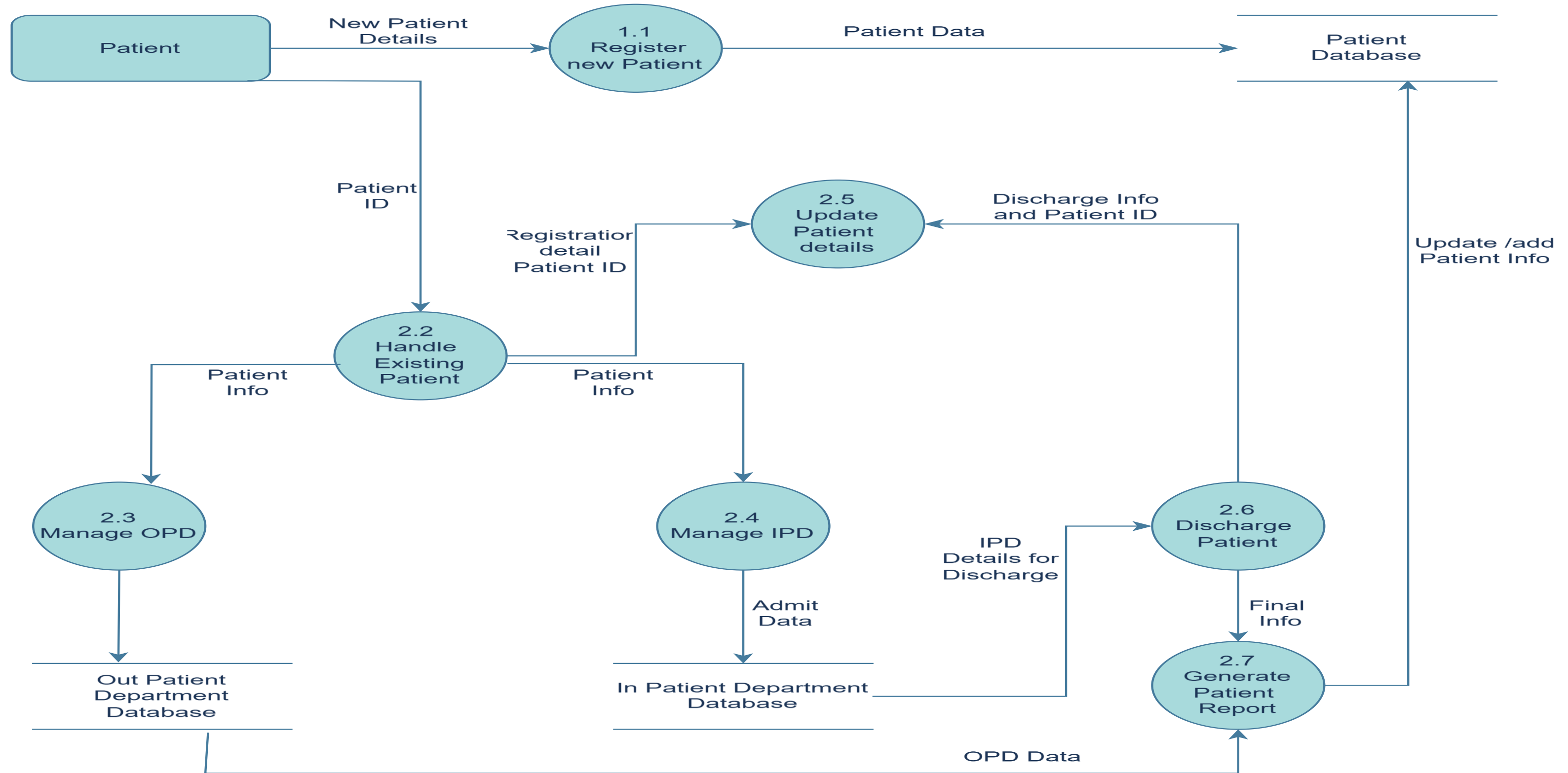
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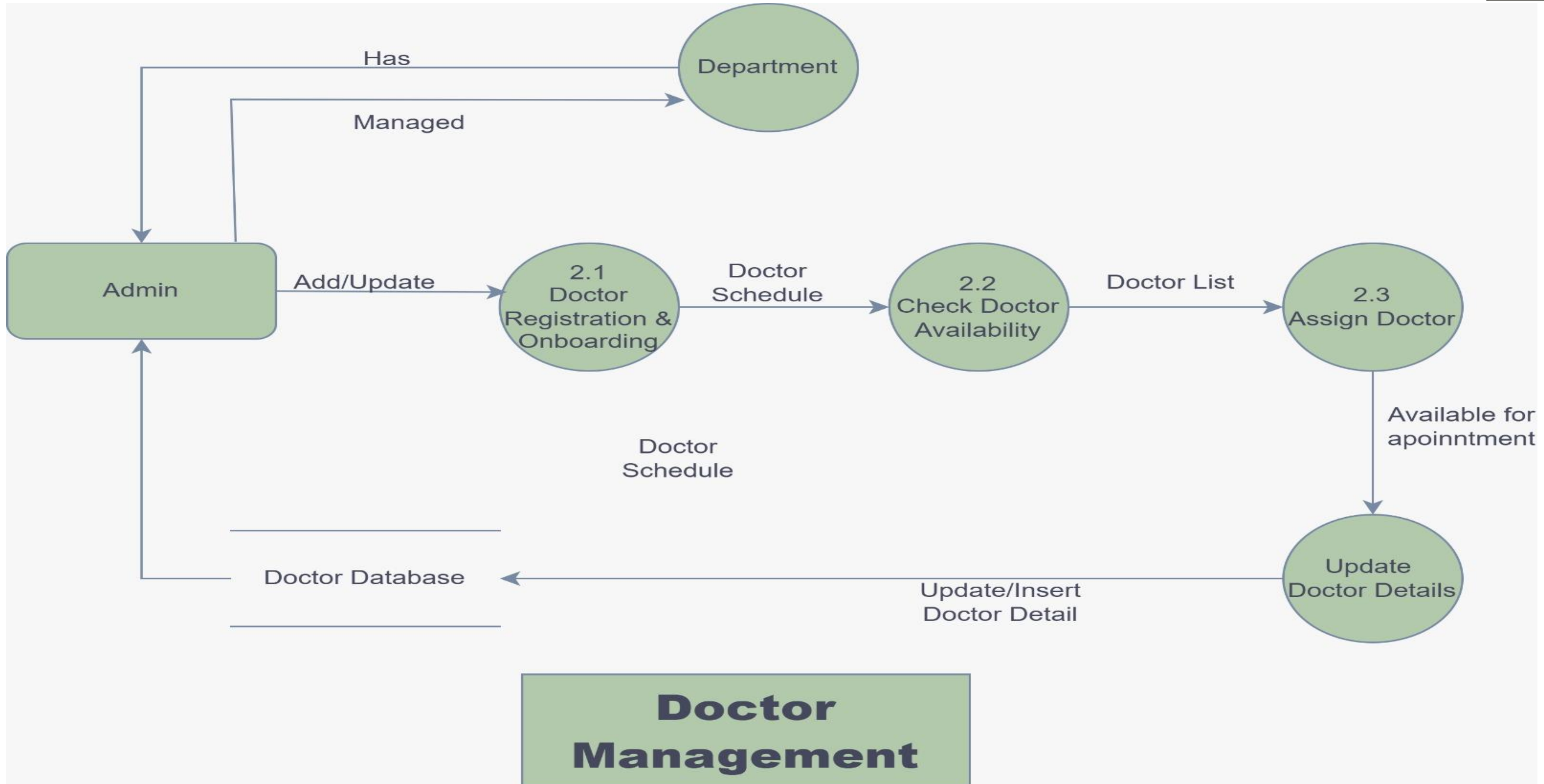


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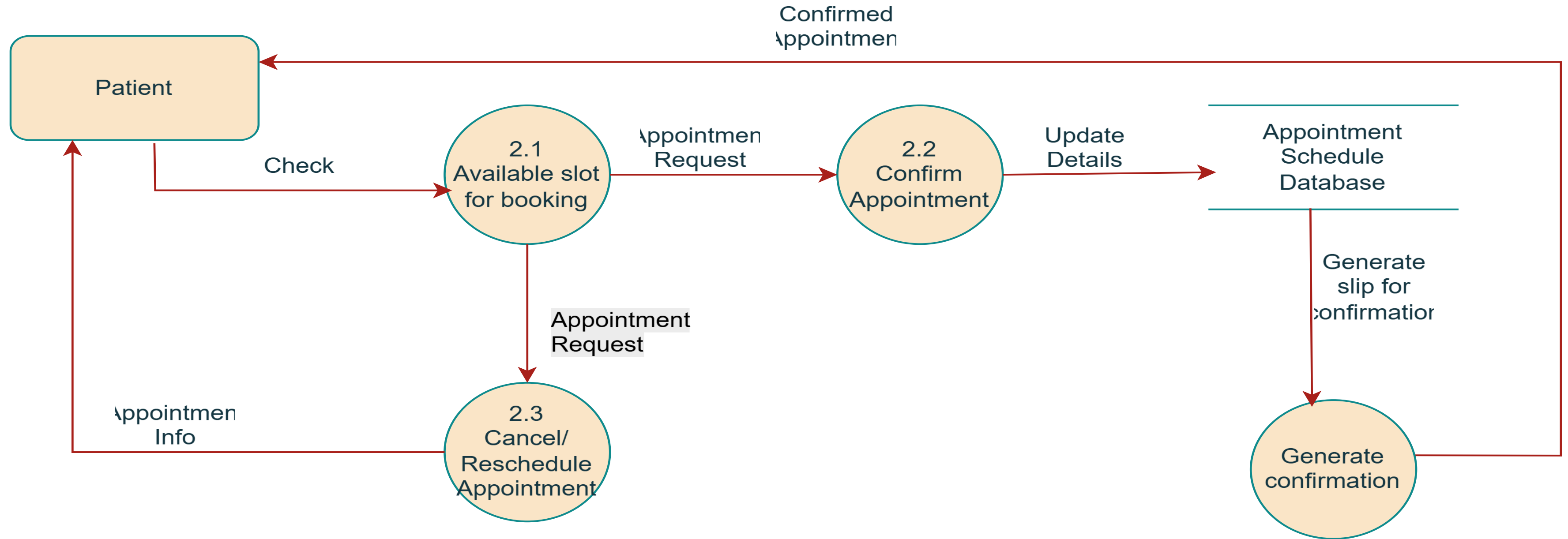


Patient Management

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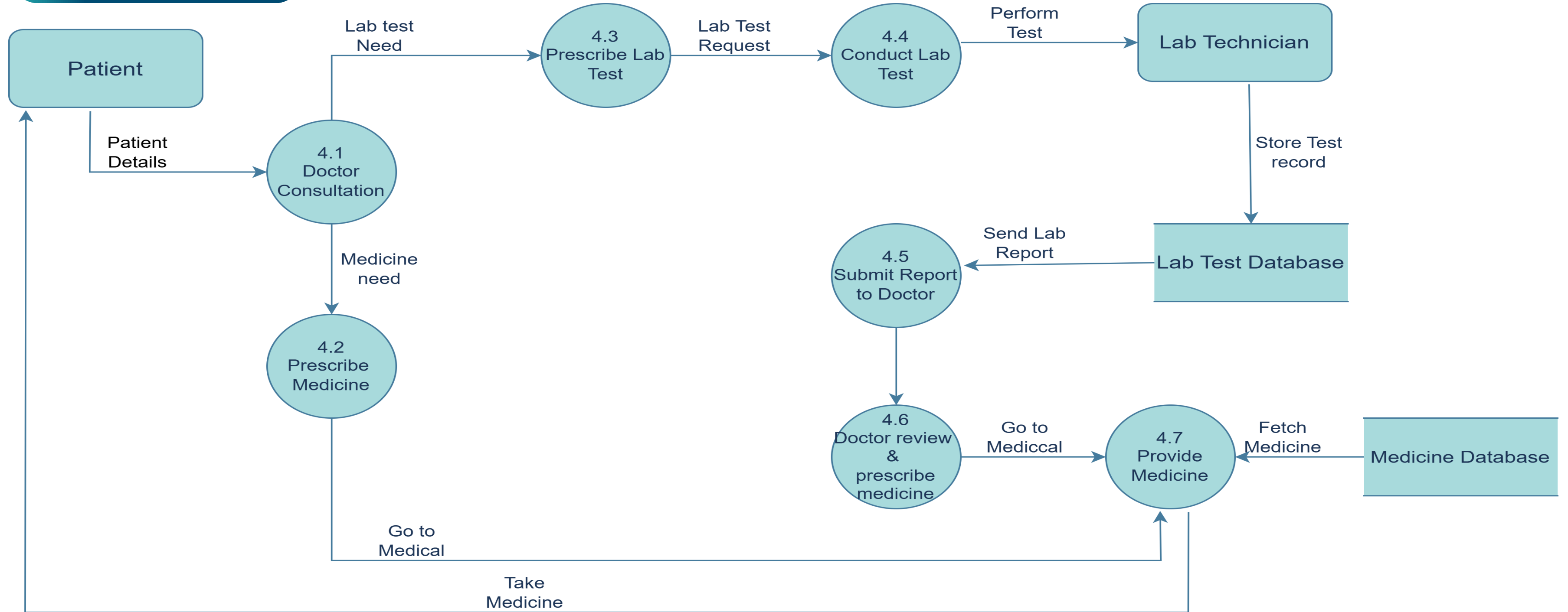


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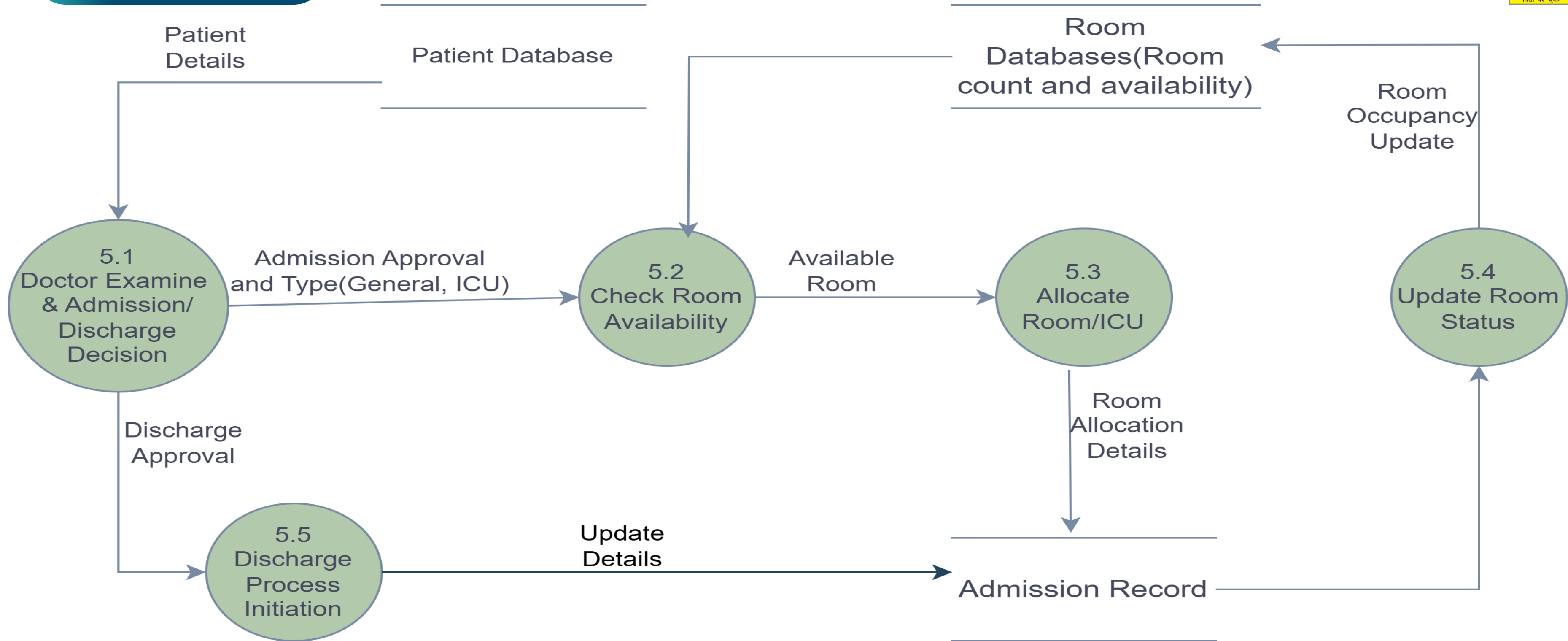
Appointment Management

DFDLEVEL-2



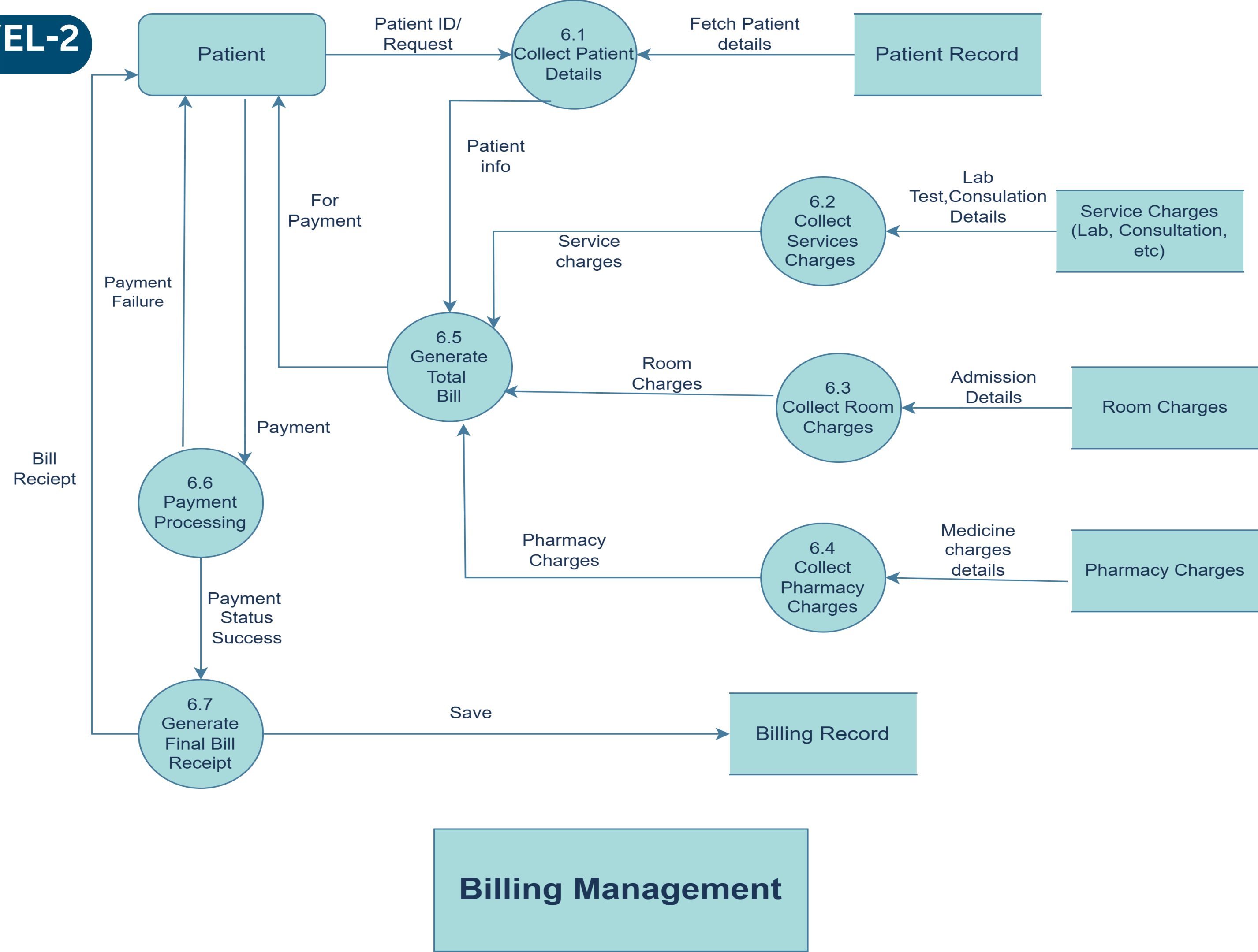
Prescription and Pharmacy

DFDLEVEL-2



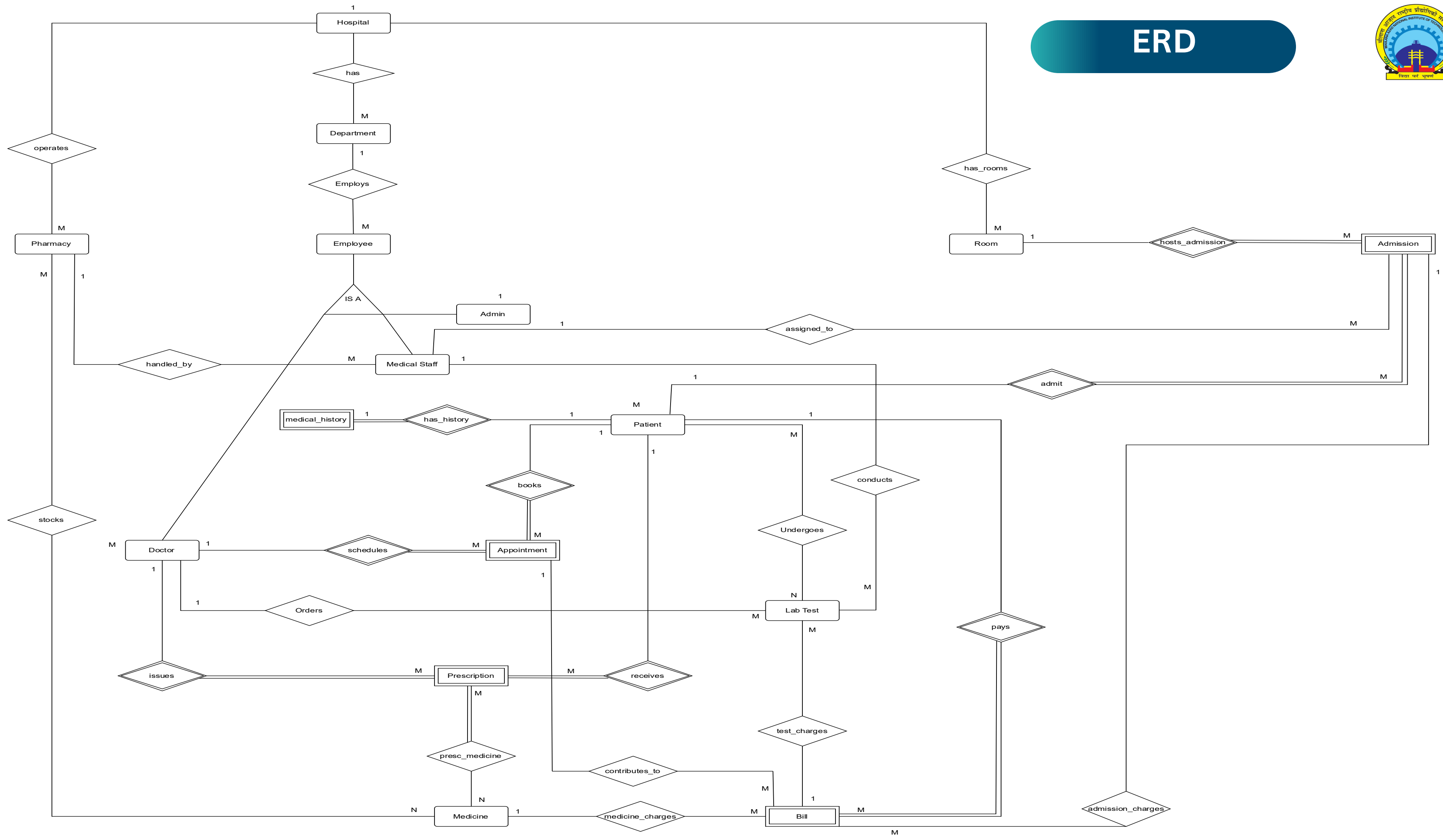
Admission And Discharge Management

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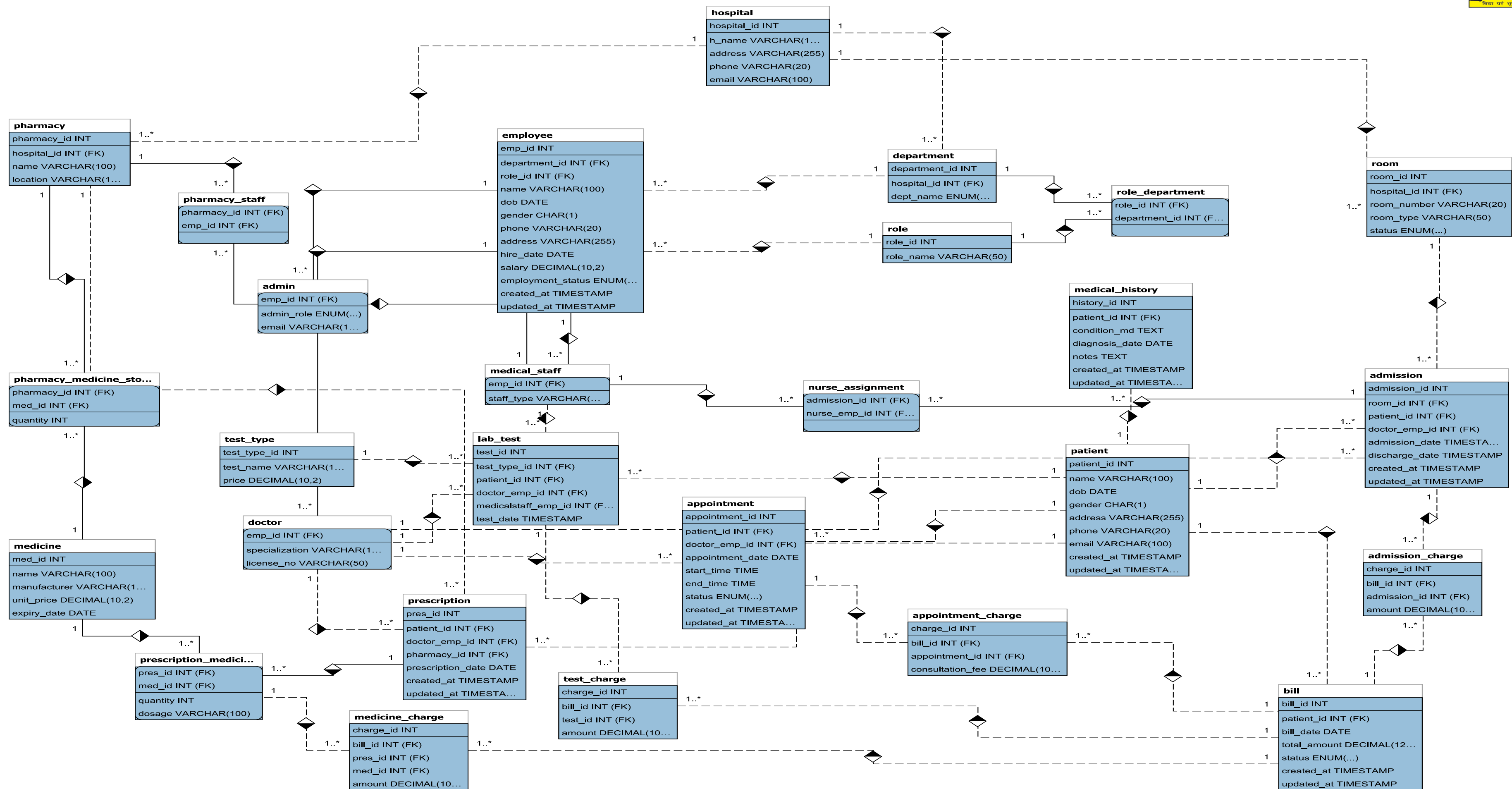


Billing Management

ERD



ERD





DATA DICTIONARY

1.Hospital

Description: The primary entity representing the medical facility's physical and contact identity.

Cardinality: 1:N with **department**, **room**, and **pharmacy**.

Field Name	Data Type	Constraints	Description
hospital_id	INT	PK, AUTO_INCREMENT	Unique identifier for the hospital.
h_name	VARCHAR(150)	NOT NULL	Full name of the hospital.
address	VARCHAR(255)	NOT NULL	Physical street address.
phone	VARCHAR(20)	CHECK (Regex)	Primary contact number.
email	VARCHAR(100)	CHECK (@)	Official contact email.

2.Department

Description: Specific medical or administrative divisions within the hospital.

Departments: Cardiology, Emergency, Pharmacy, Laboratory, Radiology, Neurology, Orthopedics, Administration.

Cardinality: 1:N with **employee**, **room**, **role_department**.

Field Name	Data Type	Constraints	Description
department_id	INT	PK, AUTO_INCREMENT	Unique ID for the department.
hospital_id	INT	FK, NOT NULL	Parent hospital reference.
dept_name	ENUM(...)	NOT NULL, UNIQUE (with hosp_id)	Restricted list of department names.

3.Role

Description: A master list of official job titles within the hospital.

Cardinality: 1:N with **employee** and **role_department**.

Field Name	Data Type	Constraints	Description
role_id	INT	PK, AUTO_INCREMENT	Unique ID for the job role.
role_name	VARCHAR(50)	UNIQUE, NOT NULL	(e.g., Doctor, Nurse, Admin).



4.Role_department

Description: A mapping table that defines which roles are permitted in specific departments.

Cardinality: M:N (Link between **role** and **department**).

Field Name	Data Type	Constraints	Description
role_id	INT	PK, FK(Role)	Reference to the role.
department_id	INT	PK, FK(department)	Reference to the department.

5.Room

Description: Physical rooms/beds available for patient admission.

Cardinality: 1:N with **admission**.

Field Name	Data Type	Constraints	Description
room_id	INT	PK, AUTO_INCREMENT	Unique ID for the room.
hospital_id	INT	FK(Hospital), NOT NULL	Hospital location of the room.
room_number	VARCHAR(20)	NOT NULL, UNIQUE (with hosp_id)	Room label/number.
room_type	VARCHAR(50)	-	Type (e.g., ICU, General Ward).
status	ENUM	DEFAULT 'available'	Current state: available, occupied, or maintenance.

6.Employee

Description: The base profile for every staff member hired by the hospital.

Cardinality: 1:1 with **doctor**, **medical_staff**, or **admin**.

Field Name	Data Type	Constraints	Description
emp_id	INT	PK, AUTO_INCREMENT	Unique internal ID for staff.
department_id	INT	FK(department), NOT NULL	The department they are assigned to.
role_id	INT	FK(Role), NOT NULL	Their specific job title ID.
name	VARCHAR(100)	NOT NULL	Full name of the employee.
dob	DATE	-	Date of birth.
gender	CHAR(1)	CHECK (M,F,O)	Biological sex or gender.
hire_date	DATE	CHECK (Not Future)	Date they started employment.
salary	DECIMAL(10,2)	CHECK (>= 0)	Base pay amount.
employment_status	ENUM	DEFAULT 'active'	active, resigned, or terminated.
created_at	TIMESTAMP	DEFAULT CURRENT	Record creation time.
updated_at	TIMESTAMP	ON UPDATE CURRENT	Last profile update.

7.Doctor

Description: Specialty attributes for doctors.

Cardinality: 1:N with [appointment](#), [prescription](#), [admission](#), and [lab_test](#).

Field Name	Data Type	Constraints	Description
emp_id	INT	PK, FK(Employee)ON DELETE CASCADE	Link to the main employee record.
specialization	VARCHAR(100)	NOT NULL	Doctor's expertise (e.g., Surgery).
license_no	VARCHAR(50)	UNIQUE, NOT NULL	Professional medical license ID.

8. Medical_staff

Description: Clinical support staff such as Nurses, Lab Technicians, and Pharmacists.

Cardinality: 1:N with [lab_test](#), [nurse_assignment](#), and [pharmacy_staff](#).

Field Name	Data Type	Constraints	Description
emp_id	INT	PK, FK(Employee)ON DELETE CASCADE	Link to the main employee record.
staff_type	VARCHAR(30)	NOT NULL	Specific clinical role.

9.Admin

Description: Specific roles for non-clinical management staff.

Cardinality: 1:1 extension of **employee**.

Field Name	Data Type	Constraints	Description
emp_id	INT	PK, FK(Employee)ON DELETE CASCADE	Link to the main employee record. 
admin_role	ENUM	NOT NULL	Specific role (e.g., HR, IT, Billing, Receptionist).
email	VARCHAR(100)	-	Work email for the admin staff.

10. Patient

Description:Master record for all patients.

Cardinality: 1:N with **appointment**, **admission**, **bill**, and **history**.

Field Name	Data Type	Constraints	Description
patient_id	INT	PK, AUTO_INCREMENT	Unique identifier for the patient.
name	VARCHAR(100)	NOT NULL	Patient's full name.
dob	DATE	-	Date of birth.
gender	CHAR(1)	CHECK (M,F,O)	Gender marker.
phone	VARCHAR(20)	CHECK (Regex)	Primary contact number.
email	VARCHAR(100)	-	Personal email address.

11. Medical_history

Description: A longitudinal record of a patient's conditions and past diagnoses.

Cardinality: N:1 with **patient**

Field Name	Data Type	Constraints	Description
history_id	INT	PK, AUTO_INCREMENT	Unique ID for the history record.
patient_id	INT	FK(Patient), NOT NULL	Patient this history belongs to.
condition_md	TEXT	-	Summary of the medical condition.
diagnosis_date	DATE	-	<u>Date the</u> condition was diagnosed.
notes	TEXT	-	Additional clinical notes.

12.Appointment

Description: Scheduled sessions for consultation between patients and doctors.

Cardinality: 1:1 with **appointment_charge**.

Field Name	Data Type	Constraints	Description
appointment_id	INT	PK, AUTO_INCREMENT	Unique ID for the visit.
patient_id	INT	FK(patient), NOT NULL	Visiting patient
doctor_emp_id	INT	FK(doctor), NOT NULL	Consulting doctor.
appointment_date	DATE	NOT NULL	Scheduled date.
start_time	TIME	NOT NULL	Start of slot.
end_time	TIME	NOT NULL	End of slot (Duration = 30m).
status	ENUM	DEFAULT 'scheduled'	scheduled, completed, or cancelled.

13. Admission

Description: Records of patient stays in hospital rooms for treatment.

Cardinality: 1:1 with admission_charge.

Field Name	Data Type	Constraints	Description
admission_id	INT	PK, AUTO_INCREMENT	Unique ID for the stay.
room_id	INT	FK(room)	Assigned room.
patient_id	INT	FK(patient)	Admitted patient.
doctor_emp_id	INT	FK(doctor)	Attending doctor.
admission_date	TIMESTAMP	-	Time of check-in.
discharge_date	TIMESTAMP	CHECK (> Admission)	Time of check-out.

14.Nurse_assignment

Description:Link between nurses and specific inpatient stays.

Cardinality: M:N (Link between admission and medical_staff).

Field Name	Data Type	Constraints	Description
admission_id	INT	PK, FK(admission)	The <u>stay being</u> monitored.
nurse_emp_id	INT	PK, FK(medical_staff)	The nurse assigned to the stay.

15.Test_type

Description: Catalog of available lab tests and their standard prices.

Cardinality: 1:N with [lab_test](#).

Field Name	Data Type	Constraints	Description
test_type_id	INT	PK, AUTO_INCREMENT	Unique ID for the test category.
test_name	VARCHAR(100)	UNIQUE	Name (e.g., Blood Test).
price	DECIMAL(10,2)	-	Base cost of the test.

16.Lab_test

Description: Results and details of tests performed on patients.

Cardinality: 1:1 with [test_charge](#).

Field Name	Data Type	Constraints	Description
test_id	INT	PK, AUTO_INCREMENT	Unique ID for the specific test.
test_type_id	INT	FK(test_type), NOT NULL	Type of test performed.
patient_id	INT	FK(patient), NOT NULL	References patient .
doctor_emp_id	INT	FK(doctor), NOT NULL	Ordering doctor.
medicalstaff_emp_id	INT	FK(medical_staff), NOT NULL	The Lab Tech who did the test.
test_date	TIMESTAMP	-	Date/Time of test.

17. Medicine

Description: Inventory list of drugs, including costs and safety dates.

Cardinality: 1:N with [pharmacy_medicine_stock](#) and [prescription_medicine](#).

Field Name	Data Type	Constraints	Description
med_id	INT	PK, AUTO_INCREMENT	Unique identifier for the drug.
name	VARCHAR(100)	NOT NULL	Drug name.
manufacturer	VARCHAR(100)	-	Producing company.
unit_price	DECIMAL(10,2)	NOT NULL, CHECK (> 0)	Price per unit.
expiry_date	DATE	NOT NULL	Safety expiration date.

18. Pharmacy

Description: physical medicine dispensary locations within the hospital.

Cardinality: 1:N with [pharmacy_staff](#) and [pharmacy_medicine_stock](#).

Field Name	Data Type	Constraints	Description
pharmacy_id	INT	PK, AUTO_INCREMENT	Unique ID for the pharmacy branch.
hospital_id	INT	FK(Hospital), NOT NULL	The hospital it belongs to.
name	VARCHAR(100)	-	Name of the dispensary.
location	VARCHAR(100)	-	Floor/Wing.

19.Pharmacy_staff

Description: Staff assigned to specific pharmacies.

Cardinality: M:N (Link between **pharmacy** and **medical_staff**).

Field Name	Data Type	Constraints	Description
pharmacy_id	INT	PK, FK(pharmacy)	Reference to the pharmacy.
emp_id	INT	PK, FK(employee)	Reference to the pharmacist.

20.Pharmacy_medicine_stock

Description: Current inventory levels of drugs at specific pharmacies.

Cardinality: Composite PK (**pharmacy_id**, **med_id**).

Field Name	Data Type	Constraints	Description
pharmacy_id	INT	PK, FK(pharmacy)	Location of the stock.
med_id	INT	PK, FK(medicine)	The specific medicine.
quantity	INT	DEFAULT 0,check(>=0)	Units in stock.

21.Prescription

Description: A doctor's official order for patient medication.

Cardinality: 1:N with [prescription_medicine](#).

Field Name	Data Type	Constraints	Description
pres_id	INT	PK, AUTO_INCREMENT	Unique ID for the prescription.
patient_id	INT	FK(patient)	Recipient patient.
doctor_emp_id	INT	FK(doctor)	The doctor who wrote it.
pharmacy_id	INT	FK(pharmacy)	Pharmacy filling the order.
prescription_date	DATE	-	Date written.

22.Prescription_medicine

Description: Detailed drug list and dosage for a specific prescription.

Cardinality: 1:1 with [medicine_charge](#).

Field Name	Data Type	Constraints	Description
pres_id	INT	PK, FK(prescription)	The parent prescription ID.
med_id	INT	PK, FK(medicine)	The specific medicine ID.
quantity	INT	NOT NULL,CHECK (> 0)	Number of units prescribed.
dosage	VARCHAR(100)	-	Frequency (e.g., 1x day).

23.Bill

Description: The final receipt for a patient's treatment.

Cardinality: 1:N with all charge tables.

Field Name	Data Type	Constraints	Description
bill_id	INT	PK, AUTO_INCREMENT	Unique invoice ID.
patient_id	INT	FK(patient), NOT NULL	Responsible patient.
bill_date	DATE	NOT NULL	Date bill generated.
total_amount	DECIMAL(12,2)	DEFAULT 0, CHECK (>=0)	Auto-sum of all charges.
status	ENUM(...)	DEFAULT 'pending'	pending or paid.

24.Appointment_charge

Description: Consultation fee for a doctor's visit added to the bill.

Cardinality: N:1 with **bill**.

Field Name	Data Type	Constraints	Description
charge_id	INT	PK, AUTO_INCREMENT	Line item ID.
bill_id	INT	FK(bill), NOT NULL	Linked master bill.
appointment_id	INT	FK(appointment), NOT NULL	Linked appointment.
consultation_fee	DECIMAL(10,2)	NOT NULL	Fixed fee for visit.

25.Medicine_charge

Description: Cost of pharmaceutical drugs dispensed added to the bill.

Cardinality: N:1 with **bill**.

Field Name	Data Type	Constraints	Description
charge_id	INT	PK, AUTO_INCREMENT	Line item ID.
bill_id	INT	FK(bill), NOT NULL	Linked master bill.
pres_id	INT	FK(prescription)	Source prescription.
med_id	INT	FK(medicine)	Drug identified.
amount	DECIMAL(10,2)	NOT NULL	Cost for medicine units.

26.Test_charge

Description: Cost of diagnostic lab tests added to the bill.

Cardinality: N:1 with **bill**.

Field Name	Data Type	Constraints	Description
charge_id	INT	PK, AUTO_INCREMENT	Unique ID for the line item.
bill_id	INT	FK(bill), NOT NULL	Reference to the master bill.
test_id	INT	FK(test)	Linked event Test.
amount	DECIMAL(10,2)	NOT NULL	Cost of the lab test.

27.Admission_charge

Description: Stay-related costs (room fees) added to the bill.

Cardinality: N:1 with **bill**.

Field Name	Data Type	Constraints	Description
charge_id	INT	PK, AUTO_INCREMENT	Unique ID for the line item.
bill_id	INT	FK(bill), NOT NULL	Reference to the master bill.
adm_id	INT	FK(admission)	Linked event Admission
amount	DECIMAL(10,2)	NOT NULL	Fee based on length of stay.

TRIGGERS

Trigger Name	Table	Event	Description
trg_validate_role_department	employee	BEFORE INSERT	Ensures role matches department
trg_validate_role_department_update	employee	BEFORE UPDATE	Validates role-department on update
trg_check_hire_date	employee	BEFORE INSERT	Prevents future hire date
trg_employee_role_distribution	employee	AFTER INSERT	Auto inserts into role tables
trg_validate_medical_staff	medical_staff	BEFORE INSERT	Valid staff type check
trg_check_pharmacist	pharmacy_staff	BEFORE INSERT	Only pharmacists allowed
trg_no_past_appointment	appointment	BEFORE INSERT	Prevents past appointments
trg_doctor_schedule	appointment	BEFORE INSERT	Avoids overlapping slots
trg_limit_doctor_daily_appointments	appointment	BEFORE INSERT	Max 20/day
trg_add_appointment_charge	appointment	AFTER INSERT	Adds consultation charge
trg_check_expired_medicine	prescription_medicine	BEFORE INSERT	Blocks expired meds
trg_check_stock	prescription_medicine	BEFORE INSERT	Checks availability
trg_reduce_stock	prescription_medicine	AFTER INSERT	Reduces stock
trg_check_lab_technician	lab_test	BEFORE INSERT	Only lab tech allowed
trg_add_test_charge	lab_test	AFTER INSERT	Adds test charge
trg_single_admission	admission	BEFORE INSERT	Single active admission
trg_room_available	admission	BEFORE INSERT	Room must be available
trg_room_occupied	admission	AFTER INSERT	Marks room occupied

Trigger Name	Table	Event	Description
trg_validate_discharge	admission	BEFORE UPDATE	Valid discharge date
trg_free_room	admission	AFTER UPDATE	Frees room
trg_auto_assign_nurse	admission	AFTER INSERT	Assigns nurse
trg_create_bill_after_admission	admission	AFTER INSERT	Creates bill
trg_admission_charge	admission	AFTER UPDATE	Adds stay charge
trg_bill_medicine_insert	medicine_charge	AFTER INSERT	Recalculate bill
trg_bill_medicine_update	medicine_charge	AFTER UPDATE	Recalculate bill
trg_bill_medicine_delete	medicine_charge	AFTER DELETE	Recalculate bill
trg_bill_test_insert	test_charge	AFTER INSERT	Recalculate bill
trg_bill_test_update	test_charge	AFTER UPDATE	Recalculate bill
trg_bill_test_delete	test_charge	AFTER DELETE	Recalculate bill
trg_bill_appt_insert	appointment_charge	AFTER INSERT	Recalculate bill
trg_bill_appt_update	appointment_charge	AFTER UPDATE	Recalculate bill
trg_bill_appt_delete	appointment_charge	AFTER DELETE	Recalculate bill
trg_bill_admission_insert	admission_charge	AFTER INSERT	Recalculate bill
trg_bill_admission_update	admission_charge	AFTER UPDATE	Recalculate bill
trg_bill_admission_delete	admission_charge	AFTER DELETE	Recalculate bill



THANK YOU